

Homework 5

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Since $SU(2) \cong S^3$, we have $H_{\text{dR}}^\bullet(SU(2)) \cong H_{\text{dR}}^\bullet(S^3)$. Let N, S denote the north and south poles of S^3 , respectively. Stereographic projection yields diffeomorphisms $S^3 \setminus \{N\} \cong \mathbb{R}^3$ and $S^3 \setminus \{S\} \cong \mathbb{R}^3$. Since $\mathbb{R}^3$ is contractible, the de Rham cohomology of $S^3 \setminus \{N\}$ and $S^3 \setminus \{S\}$ is simply $\mathbb{R}$ concentrated in degree 0 by Poincaré's lemma. Let $U := S^3 \setminus \{N\}$ and $V := S^3 \setminus \{S\}$, and note that this forms an open cover of S^3 . De Rham's theorem says that de Rham cohomology is isomorphic to singular cohomology, so we can apply the Mayer-Vietoris sequence for cohomology to obtain the long exact sequence

$$\cdots \longrightarrow H^k(S^3) \longrightarrow H^k(U) \oplus H^k(V) \longrightarrow H^k(U \cap V) \longrightarrow H^{k+1}(S^3) \longrightarrow \cdots$$

Note that $U \cap V$ has the homotopy type of the 2-sphere S^2 . We computed in class that the cohomology of the 2-sphere is

$$H_{\text{dR}}^k(S^2) \cong \begin{cases} \mathbb{R}, & k = 0, 2 \\ 0, & k \neq 0, 2 \end{cases}.$$

Since the de Rham complex functor preserves homotopy, and cohomology identifies homotopic maps, we can use the natural isomorphism $H^k(U \cap V) \cong H^k(S^2)$. We use exactness of the sequence to compute $H^k(S^3)$ for all k . Since de Rham cohomology is 0 for both S^2 and S^3 above degree 3 (for S^3 this is since it's a 3-manifold and de Rham vanishes above the dimension), and since $H^k(U) \cong H^k(V)$ is also 0 above degree 1, we only need to consider degrees $0 \leq k \leq 3$.

We first consider degree 0:

$$0 \longrightarrow H^0(S^3) \longrightarrow H^0(U) \oplus H^0(V) \longrightarrow H^0(U \cap V) \longrightarrow \cdots$$

Since $H^0(U) \cong H^0(V) \cong \mathbb{R}$, since $H^1(S^3) \cong 0$, and since $H^0(U \cap V) \cong H^0(S^2) \cong \mathbb{R}$, this reads

$$0 \longrightarrow H^0(S^3) \longrightarrow \mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow \cdots$$

By exactness, $\text{Ker}(H^0(S^3) \rightarrow \mathbb{R}^2) \cong \text{Ker}(0 \rightarrow H^0(S^3)) \cong 0$. Thus this map is injective, and $H^0(S^3)$ is canonically isomorphic to its image in $\mathbb{R}^2$. By exactness, this image is isomorphic to $\text{Ker}(\mathbb{R}^2 \rightarrow \mathbb{R})$. The map $H^k(U) \oplus H^k(V) \rightarrow H^k(U \cap V)$ in the Mayer-Vietoris sequence is given by the difference $(u, v) \mapsto u - v$, which is not identically zero. Thus it must have rank 1, and is surjective. Thus its kernel is $\mathbb{R}$ by the rank-nullity theorem, so $H^0(S^3) \cong \mathbb{R}$.

We now consider $H^1(S^3)$:

$$\cdots \longrightarrow H^0(U) \oplus H^0(V) \longrightarrow H^0(U \cap V) \longrightarrow H^1(S^3) \longrightarrow H^1(U) \oplus H^1(V) \longrightarrow \cdots$$

We plug in

$$\cdots \longrightarrow \mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow H^1(S^3) \longrightarrow 0 \longrightarrow \cdots$$

Note that $\text{Ker}(H^1(S^3) \rightarrow 0) \cong H^1(S^3)$. Exactness tells us that

$$\text{Im}(\mathbb{R} \rightarrow H^1(S^3)) \cong \text{Ker}(H^1(S^3) \rightarrow 0) \cong H^1(S^3).$$

Since $\mathbb{R}^2 \rightarrow \mathbb{R}$ is surjective, the kernel of $\mathbb{R} \rightarrow H^1(S^1)$ is the image of $\mathbb{R}^2 \rightarrow \mathbb{R}$, which is $\mathbb{R}$. Thus we obtain the zero morphism $0 : \mathbb{R} \rightarrow H^1(S^3)$. The image of this is 0, so $H^1(S^3) \cong 0$.

We now consider $H^2(S^3)$. Consider

$$\cdots \longrightarrow H^1(U \cap V) \longrightarrow H^2(S^3) \longrightarrow H^2(U) \oplus H^2(V) \longrightarrow \cdots.$$

We plug in

$$\cdots \longrightarrow 0 \longrightarrow H^2(S^3) \longrightarrow 0 \longrightarrow \cdots.$$

Exactness tells us that

$$0 \cong \text{Im}(0 \rightarrow H^2(S^3)) \cong \text{Ker}(H^2(S^3) \rightarrow 0) \cong H^2(S^3).$$

We now consider $H^3(S^3)$. The relevant part of the sequence is

$$\cdots \longrightarrow H^2(U) \oplus H^2(V) \longrightarrow H^2(U \cap V) \longrightarrow H^3(S^3) \longrightarrow H^3(U) \oplus H^3(V) \longrightarrow \cdots.$$

We plug in

$$\cdots \longrightarrow 0 \longrightarrow \mathbb{R} \longrightarrow H^3(S^3) \longrightarrow 0 \longrightarrow \cdots.$$

This is exact if and only if the map $\mathbb{R} \rightarrow H^3(S^3)$ is an isomorphism. Hence

$$H^k(S^3) \cong \begin{cases} \mathbb{R}, & k = 0, 3 \\ 0, & k \neq 0, 3 \end{cases}.$$

Recall once again that higher cohomology groups vanish since de Rham cohomology is 0 above the dimension of the manifold.

We claim there exists a basis $\{X_1, X_2, X_3\} \subseteq \mathfrak{su}(2)$ satisfying $[X_i, X_j] = \varepsilon_{ijk} X_k$, for ε the Levi-Civita symbol. We use the characterization

$$\mathfrak{su}(2) = \left\{ A \in \mathfrak{gl}(2, \mathbb{C}) \mid A^\dagger + A = 0 \text{ and } \text{tr } A = 0 \right\},$$

where $A^\dagger$ denotes the Hermitian conjugate. Recall that the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

satisfy the commutator relation

$$[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k.$$

Define $X_i = -\frac{i}{2}\sigma_i$. Since each Pauli matrix is traceless, so is its product by $-\frac{i}{2}$. We also have

$$X_1^\dagger = \begin{pmatrix} 0 & -i/2 \\ i/2 & 0 \end{pmatrix}^\dagger = \begin{pmatrix} 0 & i/2 \\ i/2 & 0 \end{pmatrix} = -X_1,$$

$$X_2^\dagger = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}^\dagger = \begin{pmatrix} 0 & -1/2 \\ 1/2 & 0 \end{pmatrix} = -X_2,$$

$$X_3^\dagger = \begin{pmatrix} -i/2 & 0 \\ 0 & i/2 \end{pmatrix}^\dagger = \begin{pmatrix} i/2 & 0 \\ 0 & -i/2 \end{pmatrix} = -X_3.$$

Finally,

$$[X_i, X_j] = \left(-\frac{i}{2}\right)^2 [\sigma_i, \sigma_j] = \left(-\frac{i}{2}\right)^2 2i\varepsilon_{ijk}\sigma_k = 2i\frac{-i}{2}\varepsilon_{ijk}\left(-\frac{i}{2}\sigma_k\right) = \varepsilon_{ijk}X_k.$$

Thus such a basis exists. Let $\{\alpha_1, \alpha_2, \alpha_3\}$ be the dual basis.

Recall that Lie algebra cohomology of a Lie $\mathbb{R}$ -algebra $\mathfrak{g}$ can be computed by the cohomology of the Chevalley-Eilenberg complex, which can be identified with the subcomplex of the de Rham complex $\Omega^\bullet(\mathrm{SU}(2))$ spanned by left-invariant forms. Write d for the Chevalley-Eilenberg differential, which in this case coincides with the exterior derivative. Then we compute

$$d\alpha_i(X_j, X_k) = X_j\alpha_i(X_k) - X_k\alpha_i(X_j) - \alpha_i[X_j, X_k].$$

Here note that we define left-invariant vector fields for the X_i 's. For any left-invariant vector field X , left-invariant form α , and $g \in G$,

$$\alpha_g(X_g) = (L_{g^{-1}}^*\alpha_1)((L_g)_*X_1) = \alpha_1(X_1).$$

In particular, $\alpha(X)$ is constant for any left-invariant X . Thus $X_j\alpha_i(X_k)$ vanishes for all indices, since as a vector field X_j is a linear combination of partial derivatives acting on a constant function. Thus

$$d\alpha_i(X_j, X_k) = -\alpha_i[X_j, X_k].$$

Using the commutation relation,

$$-\alpha_i[X_j, X_k] = -\alpha_i(\varepsilon_{jkl}X_l) = -\varepsilon_{jkl}\alpha_i(X_l) = -\varepsilon_{jkl}\delta_{il} = -\varepsilon_{jki} = -\varepsilon_{ijk}$$

for δ the Kronecker delta. This needs to be a 2-form, and thus must be of the form $c\alpha_j \wedge \alpha_k$. We note that if $i \neq j, k$, then

$$(c\alpha_j \wedge \alpha_k)(X_j, X_k) = c\alpha_j(X_j)\alpha_k(X_k) - c\alpha_k(X_j)\alpha_j(X_k) = c.$$

Note that if $c = -\varepsilon_{ijk}$, then we have $d\alpha_i = -\varepsilon_{ijk}\alpha_j \wedge \alpha_k$ on basis vectors, and thus they should agree in general. Thus we have computed $d\alpha_i$.

Now consider the Chevalley-Eilenberg complex for $\mathfrak{su}(2)$:

$$0 \xrightarrow{d^0} \Lambda^0 \mathfrak{su}(2)^* \xrightarrow{d^1} \mathfrak{su}(2)^* \xrightarrow{d^2} \Lambda^2 \mathfrak{su}(2)^* \xrightarrow{d^3} \Lambda^3 \mathfrak{su}(2)^* \xrightarrow{d^4} 0$$

Since $\Lambda^0 \mathfrak{su}(2)^* \cong \mathbb{R}$ is the space of constant functions, and the differential acts trivially on a constant function, $\mathrm{Ker} d^1 \cong \mathbb{R}$. Since $\mathrm{Im} d^0 \cong 0$, we have $H^0(\mathfrak{su}(2)) \cong \mathbb{R}$. Now we use the action of d^2 on the basis

$$d^2(\alpha_1) = -\alpha_2 \wedge \alpha_3, \quad d^2(\alpha_2) = -\alpha_3 \wedge \alpha_1, \quad d^2(\alpha_3) = -\alpha_1 \wedge \alpha_2.$$

Note that this is clearly a basis for $\Lambda^2 \mathfrak{su}(2)^*$, so d^2 is an isomorphism. Its kernel is therefore trivial, and since the image of d^1 is also trivial, $H^1(\mathfrak{su}(2)) \cong 0$. We also obtain that the image of d^2 is the entire vector space $\Lambda^2 \mathfrak{su}(2)^*$, so d^3 must be the zero map. The cohomology $H^2(\mathfrak{su}(2))$ must then also be 0 since the sequence is exact in degree 2. Finally, since d^3 has trivial image and d^4 is the zero morphism, $H^3(\mathfrak{su}(2)) \cong \mathrm{Ker} d^4 \cong \Lambda^3 \mathfrak{su}(2)^*$. Since $\Lambda^3 \mathfrak{su}(2)^*$ is 1-dimensional with basis $\{\alpha_1 \wedge \alpha_2 \wedge \alpha_3\}$, the third cohomology group is $\mathbb{R}$. Therefore, $H^\bullet(\mathfrak{su}(2)) \cong H_{\mathrm{dR}}^\bullet(\mathrm{SU}(2))$.